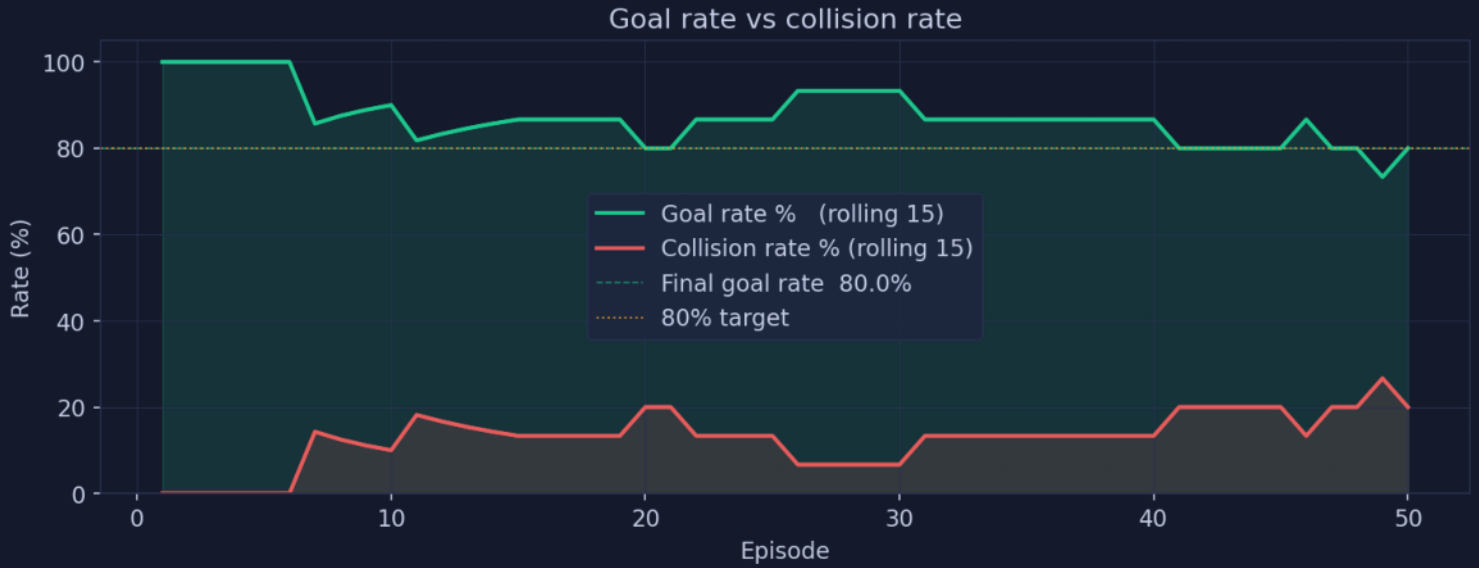
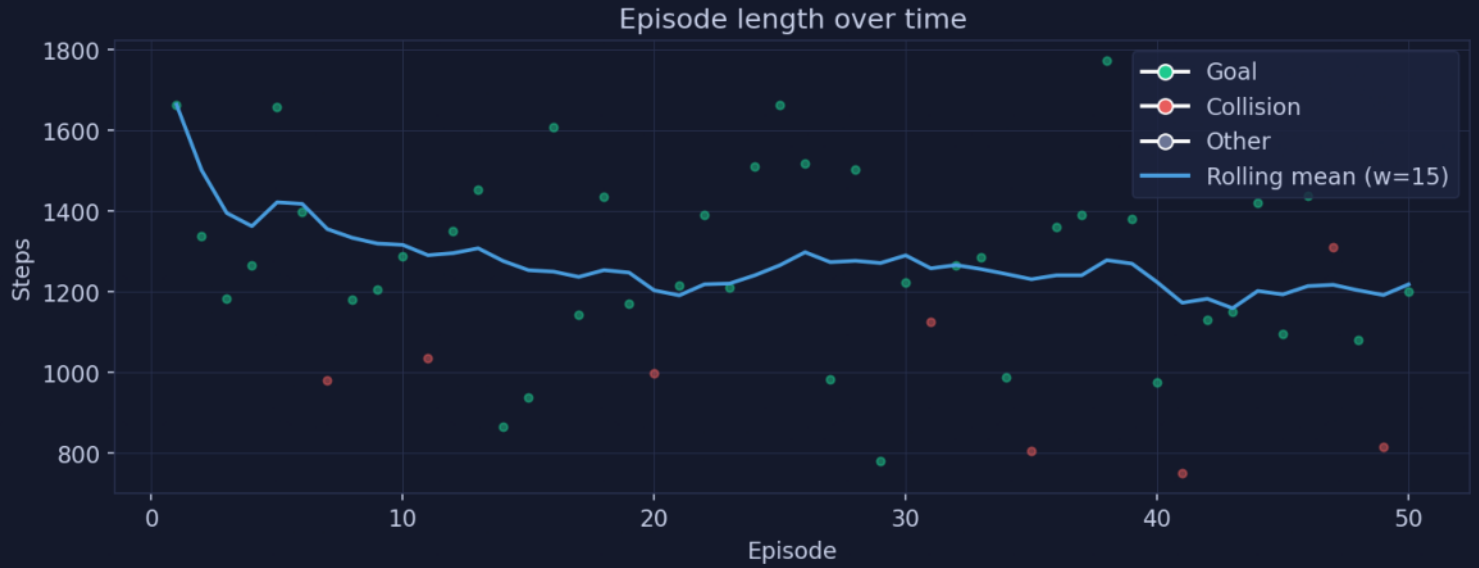
RL Maritime Autopilot

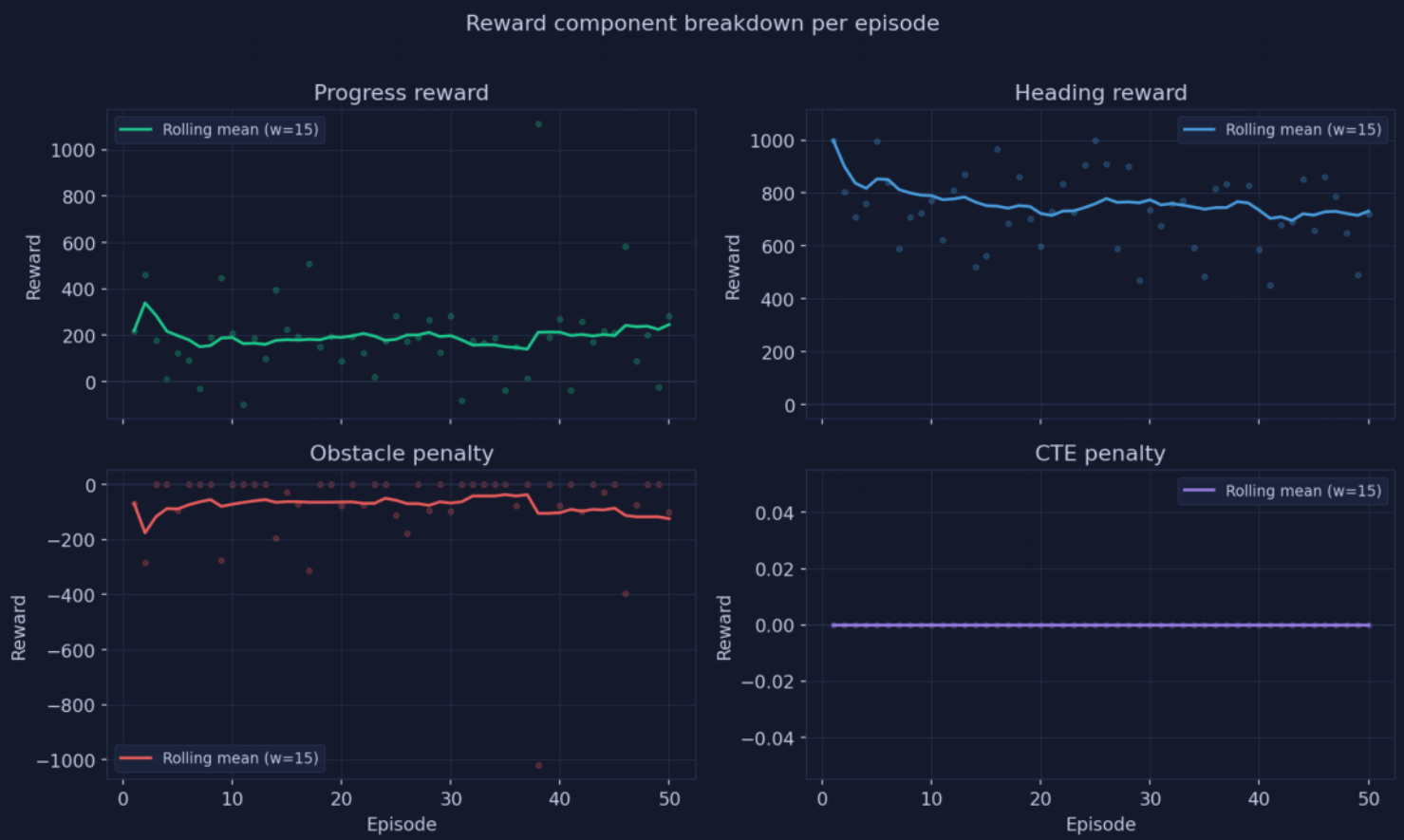
Performance Metrics Report

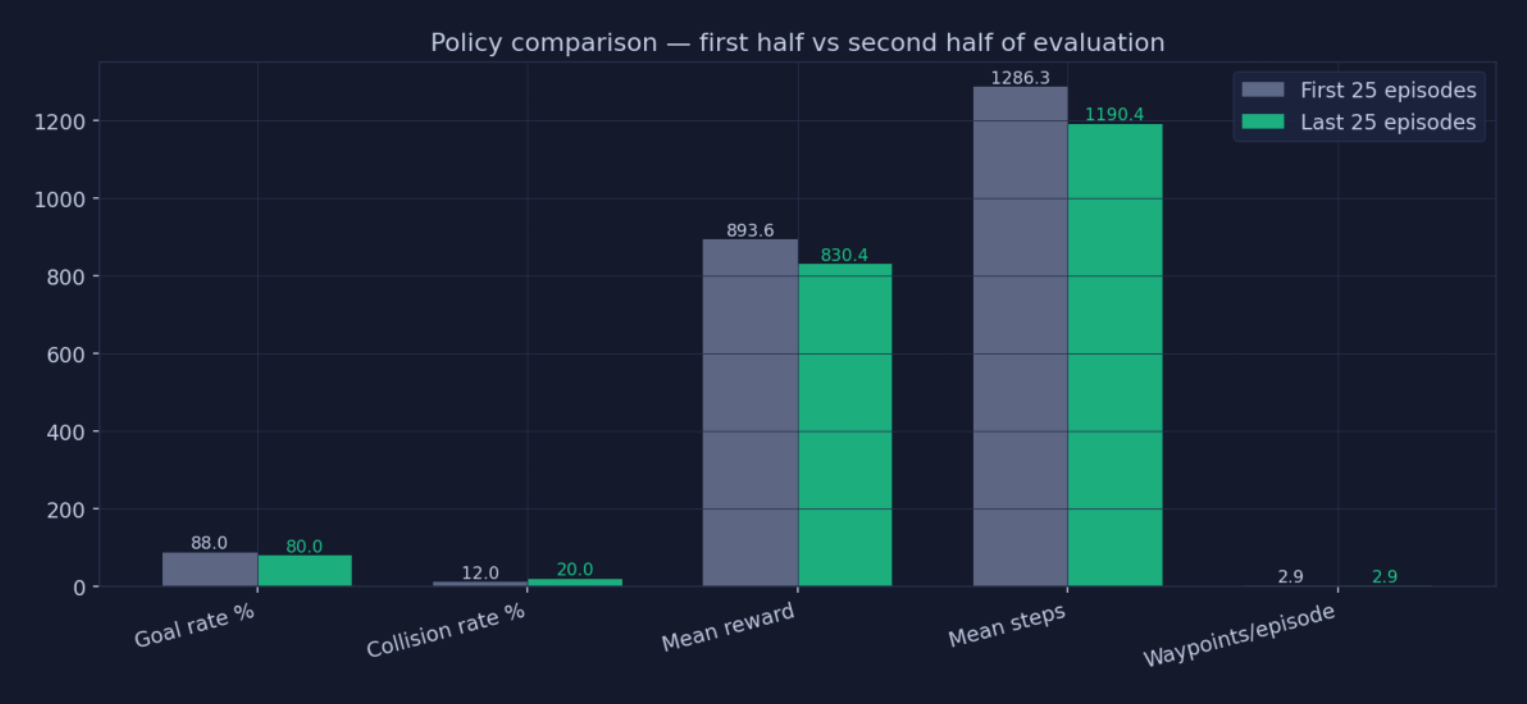
```
Total episodes evaluated : 50
Goal rate                : 84.0%
Collision rate           : 16.0%
Mean episode reward      : 862.0
Mean episode length      : 1238 steps
Mean waypoints / episode : 2.90 / 3
Rolling window used      : 15 episodes
```

Episode reward over time
Total reward
Episode
Raw reward
Rolling mean (w=15)
1200
1000
800
600
400
200
0
0
10
20
30
40
50


Goal rate vs collision rate
Rate (%)
Episode
Goal rate % (rolling 15)
Collision rate % (rolling 15)
Final goal rate 80.0%
80% target
0
20
40
60
80
100
0
10
20
30
40
50


Episode length over time
Steps
Episode
1800
1600
1400
1200
1000
800
0
10
20
30
40
50
Goal
Collision
Other
Rolling mean (w=15)


Reward component breakdown per episode
Progress reward
Rolling mean (w=15)
Reward
1000
800
600
400
200
0
Heading reward
Rolling mean (w=15)
Reward
1000
800
600
400
200
0
Obstacle penalty
Rolling mean (w=15)
Reward
0
−200
−400
−600
−800
−1000
CTE penalty
Rolling mean (w=15)
Reward
0.04
0.02
0.00
−0.02
−0.04
0
10
20
30
40
50
Episode
0
10
20
30
40
50
Episode


Policy comparison — first half vs second half of evaluation
First 25 episodes
Last 25 episodes
0
200
400
600
800
1000
1200
88.0
80.0
12.0
20.0
893.6
830.4
1286.3
1190.4
2.9
2.9
Goal rate %
Collision rate %
Mean reward
Mean steps
Waypoints/episode